ENVS193DS-homework-03

Kate Bierwagen

2026-05-28

Table of contents

Setting Up

```
# loading in packages
library(tidyverse)
library(here)
library(janitor)
library(readxl)
```

```
# reading in data
kelp <- read_csv(here("data", "temp-kelp.csv"))
my_data <-read_csv(here("data", "stats-personal-data_0523.csv"))
```

Problem 1. Giant kelp fronds

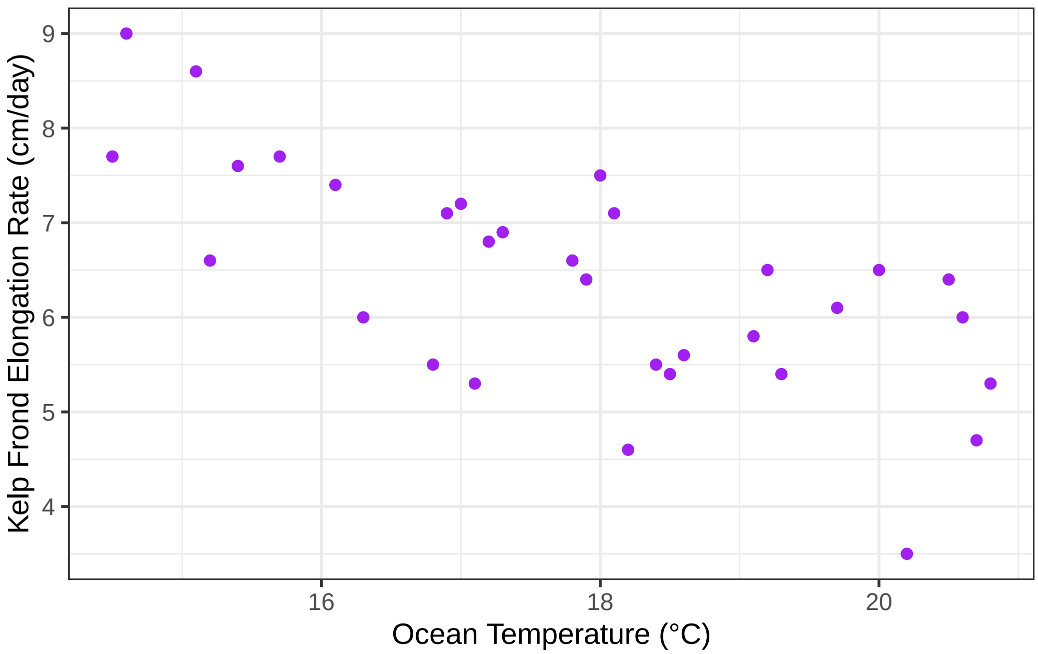
a. An appropriate test

Since you are interested in the strength of the relationship between temperature and giant kelp frond elongation rate, the appropriate test is a Pearson's r or a Spearman rank correlation, which are both tests for correlation. Pearson's r test is a parametric version of a correlation test, used for continuous and normally distributed variables, testing the linear relationship between them. Spearman rank correlation is a non-parametric version, which is testing the monotonic relationship between variables.

b. Create a visualization

```
# base layer: ggplot
ggplot(data = kelp, # using kelp data frame
       aes(x = temp_c, # x- axis: Ocean temperature (degrees C)
           y = kelp_elong)) + # y axis: kelp elongation rate (cm/day)
  # first layer: points
  geom_point(
    color = "purple") + #changing color from default
  # relabeling x- and y- axes
```

```
   labs(
       x = "Ocean Temperature (°C)", #
       y = "Kelp Frond Elongation Rate (cm/day)") +
  # changing theme from default
  theme_bw()
```



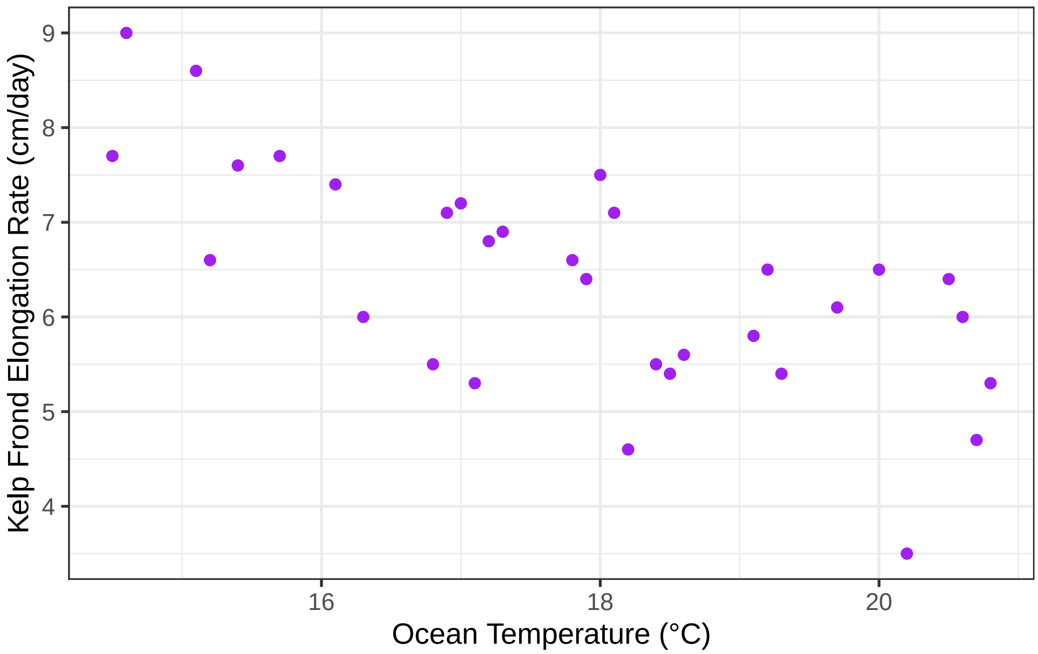
c. Check your assumptions and run your test

Checking assumptions

Check 1: Linearity

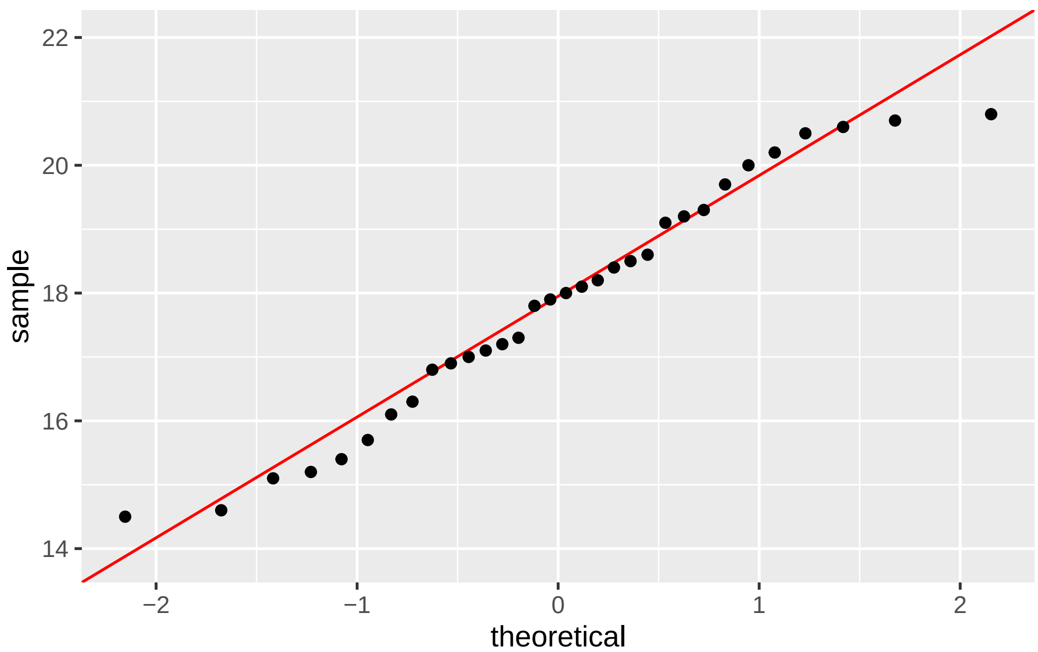
```
# base layer: ggplot
ggplot(data = kelp, # using kelp data frame
       aes(x = temp_c, # x- axis: Ocean temperature (degrees C)
           y = kelp_elong)) + # y axis: kelp elongation rate (cm/day)
  # first layer: points
  geom_point(
    color = "purple") + #changing color from default
  # relabeling x- and y- axes
   labs(
```

```
       x = "Ocean Temperature (°C)", #
       y = "Kelp Frond Elongation Rate (cm/day)") +
  # changing theme from default
  theme_bw()
```

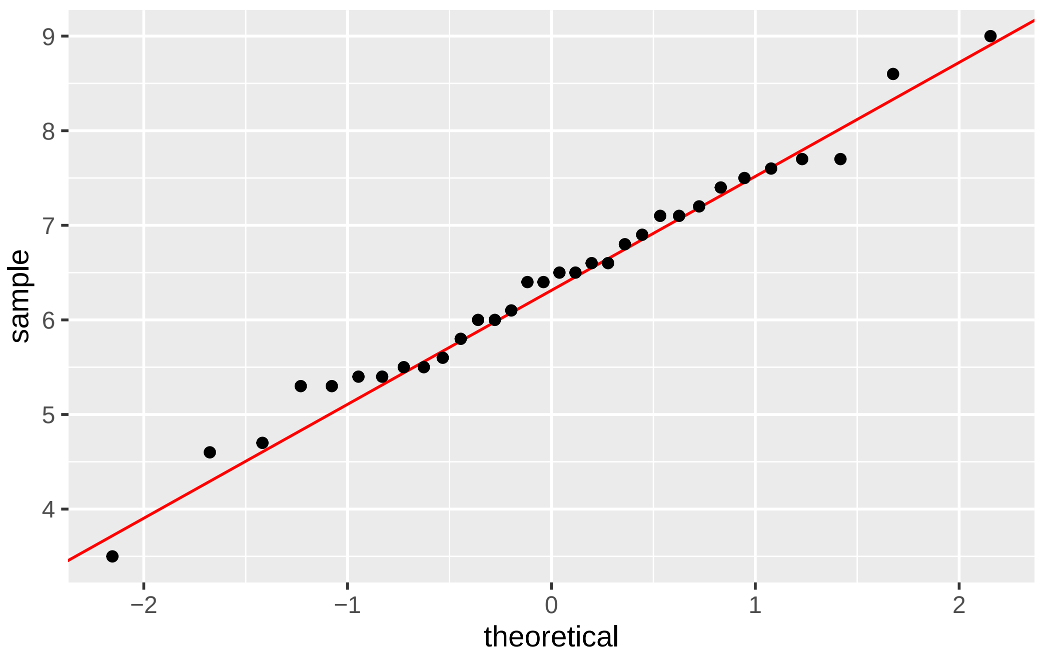


Check 2: Normality

```
# base layer: ggplot, using kelp object,
ggplot(data = kelp,
       # y axis for QQ plot, ocean temperatire
       aes(sample = temp_c)) +
  # first layer: QQ reference line
  geom_qq_line(color = "red") +
  # second layer: QQ plot
  geom_qq()
```




```
# Kelp elongation normality
# base layer: ggplot, using ukelp object
ggplot(data = kelp,
       # y axis for QQ plot (kelp frond elongation rate)
       aes(sample = kelp_elong)) +
  # first layer: QQ reference line
  geom_qq_line(color = "red") +
  # second layer: QQ plot
  geom_qq()
```



Q-Q plots were created to check that both variables, ocean temperature (°C) and kelp elongation rate (cm/day), were normally distributed. Then, a scatter plot was created to check if there was a linear or monotonic relationship between variables. Both variables are continuous variables, are normally distributed as they follow the theoretical distribution reference lines, and follow a relatively linear relationship so a Pearson's r test is recommended.

Running test:

```
# running correlation test
cor.test(kelp$temp_c, kelp$kelp_elong, # variables ocean temperature and
# giant kelp elongation rate
         method = "pearson") # running pearson's r based on assumptions
```

```

	Pearson's product-moment correlation

data:  kelp$temp_c and kelp$kelp_elong
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
```

```
sample estimates:
       cor
-0.6877489
```

d. Results communication

To evaluate the strength of the relationship between ocean temperature and giant kelp (*Macrocystis pyrifera*) frond elongation rate, I used a Pearson's correlation test. The test revealed a moderate negative relationship between ocean temperature and kelp elongation rate (Pearson's r = - 0.69, t(30) = - 5.19, p $<$ 0.001, $\alpha = 0.05$).

e. Test implications

Our results revealed a moderate negative relationship between ocean temperature and Giant Kelp Elongation rate. This means that at higher ocean temperatures, giant kelp frond elongation rate is lower, Ultimately, these results suggest that higher ocean temperatures threaten kelp health and kelp individuals should be planted in lower ocean temperatures.

f. Double check your own work

```
# running correlation test
cor.test(kelp$temp_c, kelp$kelp_elong, # variables ocean temperature and
# giant kelp elongation rate
         method = "spearman") # Spearman's rank test
```

```

    Spearman's rank correlation rho

data:  kelp$temp_c and kelp$kelp_elong
S = 9216.1, p-value = 1.29e-05
alternative hypothesis: true rho is not equal to 0
sample estimates:
       rho
-0.6891684
```

Based on the correlation coefficient from the Spearman's rank test (ρ = -0.69) and p- value (p $<$ 0.01), I would make the same decision to reject the null hypothesis, meaning that there is a statistically significant relationship between ocean temperatures and giant kelp elongation rate. I would also interpret the results the same, that there is a moderate negative relationship between ocean temperature and giant kelp elongation rate. (Spearman's ρ = -0.69, S = 9216.1, p = < 0.01, α = 0.05).

Problem 2. Personal data

a. Updating your visualizations

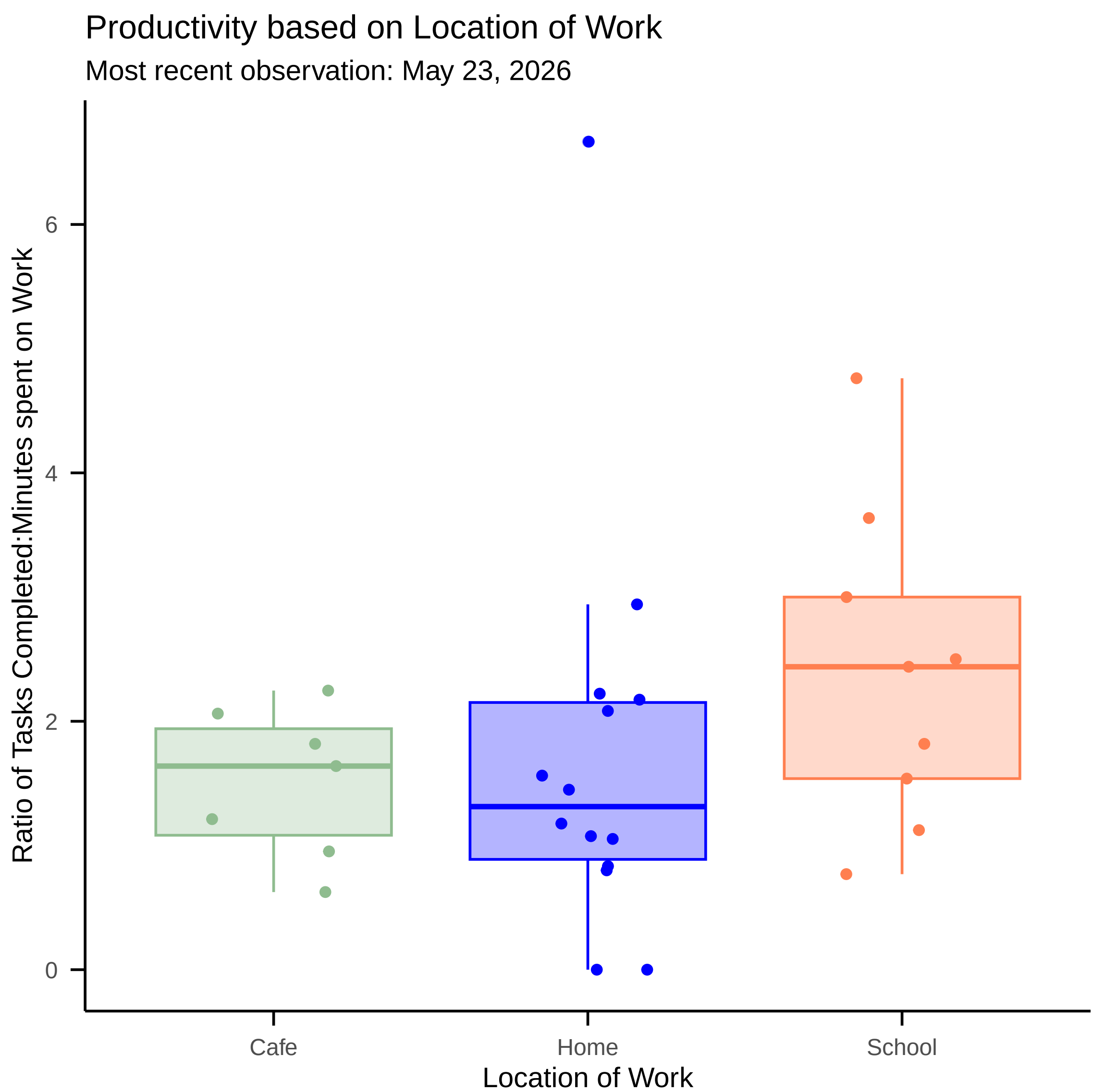
Cleaning data:

```
# creating new object from my_data
my_data_clean <- my_data |>
  # cleaning names
  clean_names() |>
  # dropping na values
  drop_na() |>
  # making productivity larger numbers, ), convert column to numeric,
  # instead of character
  mutate(productivity = as.numeric(productivity),
         productivity_hun = productivity*100)
```

Creating figures:

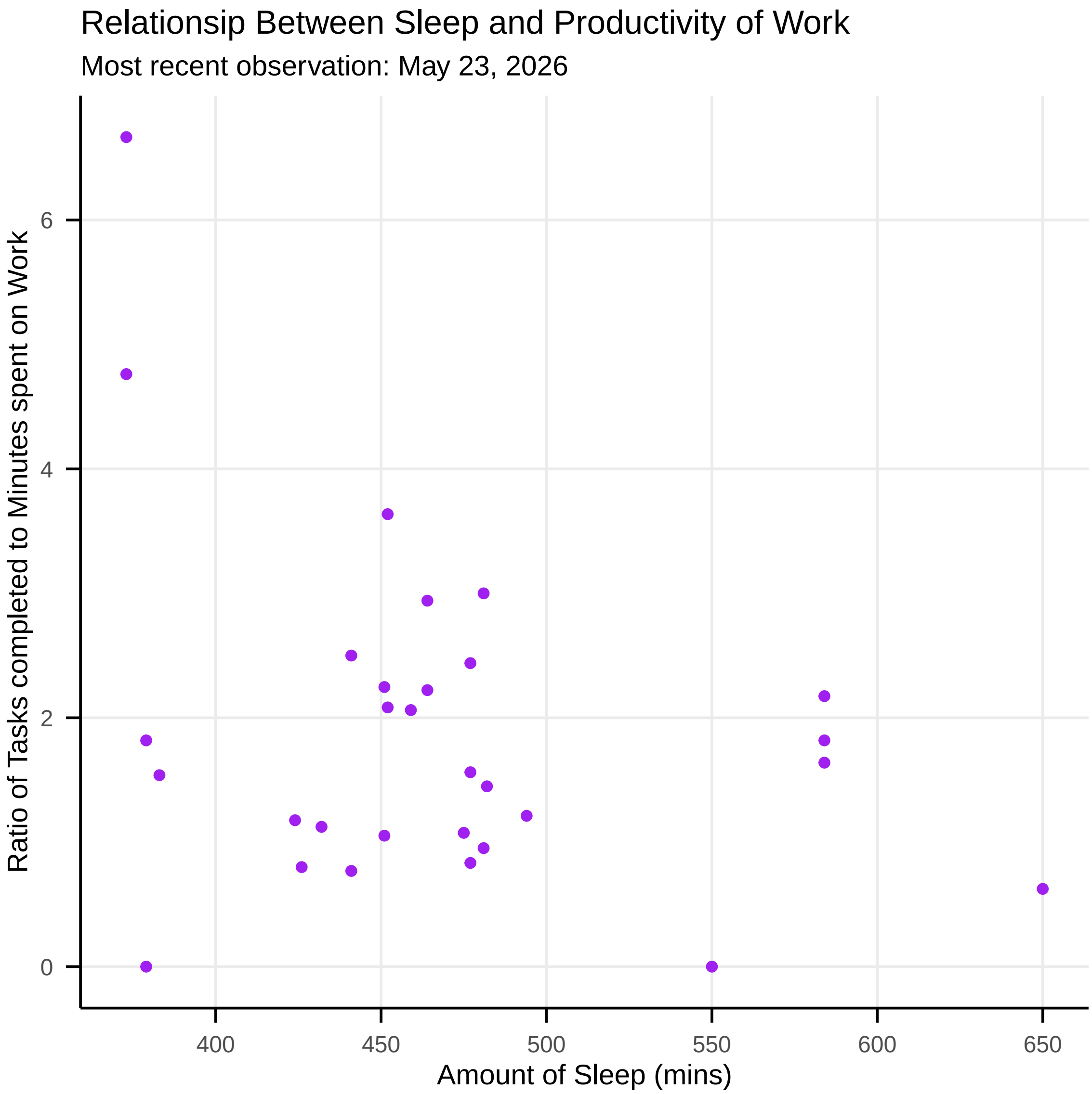
```
# base layer: ggplot using clean data frame
ggplot(data = my_data_clean,
       mapping = aes(x = location, # x-axis: location
                     y = productivity_hun, # y-axis productivity x 100
                     color = location, # coloring by location
                     fill = location)) +# filling boxplot by location
  # first layer: boxplot
  geom_boxplot(alpha = 0.3) + # making boxplot more transparent
  # second layer: showing underlying data
  geom_jitter(height = 0, # making jitter vertically
              width = 0.2, # smaller jitter in horizontal direction
              shape = 19) + # changing shape to closed circles
  # adding title and relabeling x- and y- axes
  labs(title = "Productivity based on Location of Work",
       x = "Location of Work",
       y = "Ratio of Tasks Completed:Minutes spent on Work",
```

```
     subtitle = "Most recent observation: May 23, 2026") + # adding subtitle
  # changing colors of locations from default
  scale_color_manual(values = c("home" = "blue",
                                "school" = "coral",
                                "cafe" = "darkseagreen")) +
  # changing fill colors of boxplots
  scale_fill_manual(values = c("home" = "blue",
                               "school" = "coral",
                               "cafe" = "darkseagreen")) +
  # relabelling location names to be Capitalized
  scale_x_discrete(labels = c("cafe" = "Cafe",
                                "home" = "Home",
                                "school" = "School")) +
  # changing theme from default
  theme_minimal() +
  theme(legend.position = "none", # removing legend
        panel.grid = element_blank(), # removing gridlines
         axis.line = element_line(color = "black"), # adding black axis lines
        axis.ticks = element_line(color = "black"), # adding axis ticks
        axis.ticks.length = unit(0.2, "cm")) # making tick marks 2 cm long
```




```
# base layer: ggplot using cleaned data
ggplot(data = my_data_clean,
       mapping = aes(x = sleep_mins, # x axis: sleep recorded
                     y = productivity_hun)) + # y axis: ratio of tasks
  # compeleted to minutes spent on work
  # first layer: points to display scatter plot
    geom_point(color = "purple",  # changing color of points from default
               size = 1.5) + # changing size of pints
  # adding title, relabeling x- and y- axes
  labs(title = "Relationsip Between Sleep and Productivity of Work",
```

```
     x = "Amount of Sleep (mins)",
     y = "Ratio of Tasks completed to Minutes spent on Work",
     subtitle = "Most recent observation: May 23, 2026") + # adding subtitle
scale_x_continuous(breaks = seq(300, 700, by = 50)) + # setting x axis ticks
# every 50 mins
scale_y_continuous(breaks = seq(0, 8, by = 2)) + # setting y axis ticks every
# 2 ratio to reduce visual clutter
# changing theme from default
theme_minimal() +
theme(panel.grid.minor = element_blank(),# removing minor grid lines to
# reduce visual clutter
      axis.line = element_line(color = "black"), # adding black axis lines
      axis.ticks = element_line(color = "black"), # adding black axis ticks
      axis.ticks.length = unit(0.2, "cm")) # making tick marks 2 cm long
```



b. Captions

Figure 1. School displayed a higher productivity level compared to cafe and home locations. Points represent individual work sessions (n = 30, most recent observation May 23, 2026). Boxes represent IQR and lines represent medians. School location (orange) showed highest median ratio of tasks completed to minutes spent on work with the widest interquartile range.

Figure 2. More sleep does not reliably lead to higher productivity. Data points

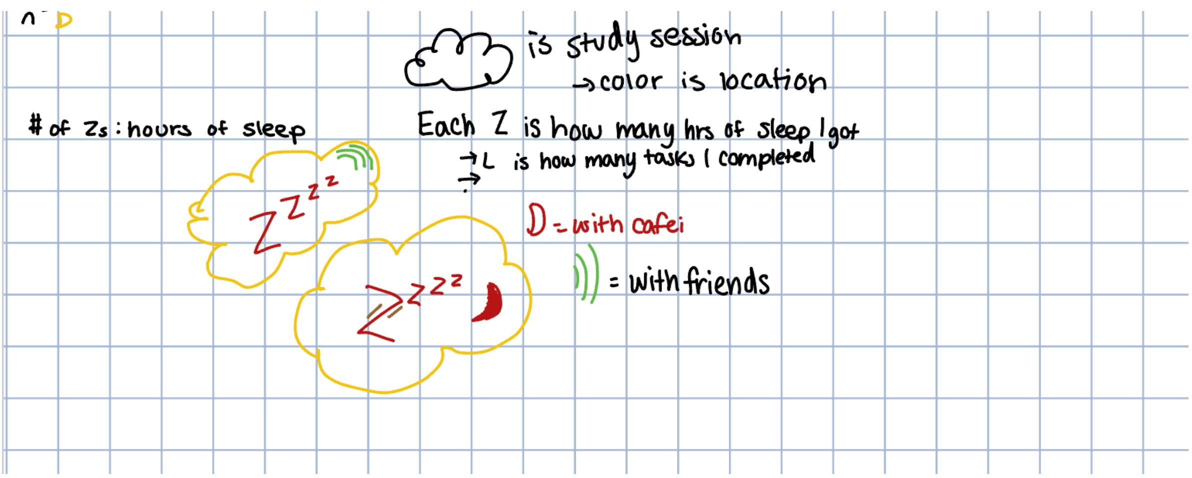
in purple represent individual work sessions (n = 30, most recent observation May 23, 2026), with no consistent upward trend observed as sleep duration increases. The highest recorded ratio of tasks completed to minutes spent on work occurred with the one of the lowest sleep duration (373 minutes).

Problem 3. Affective visualization

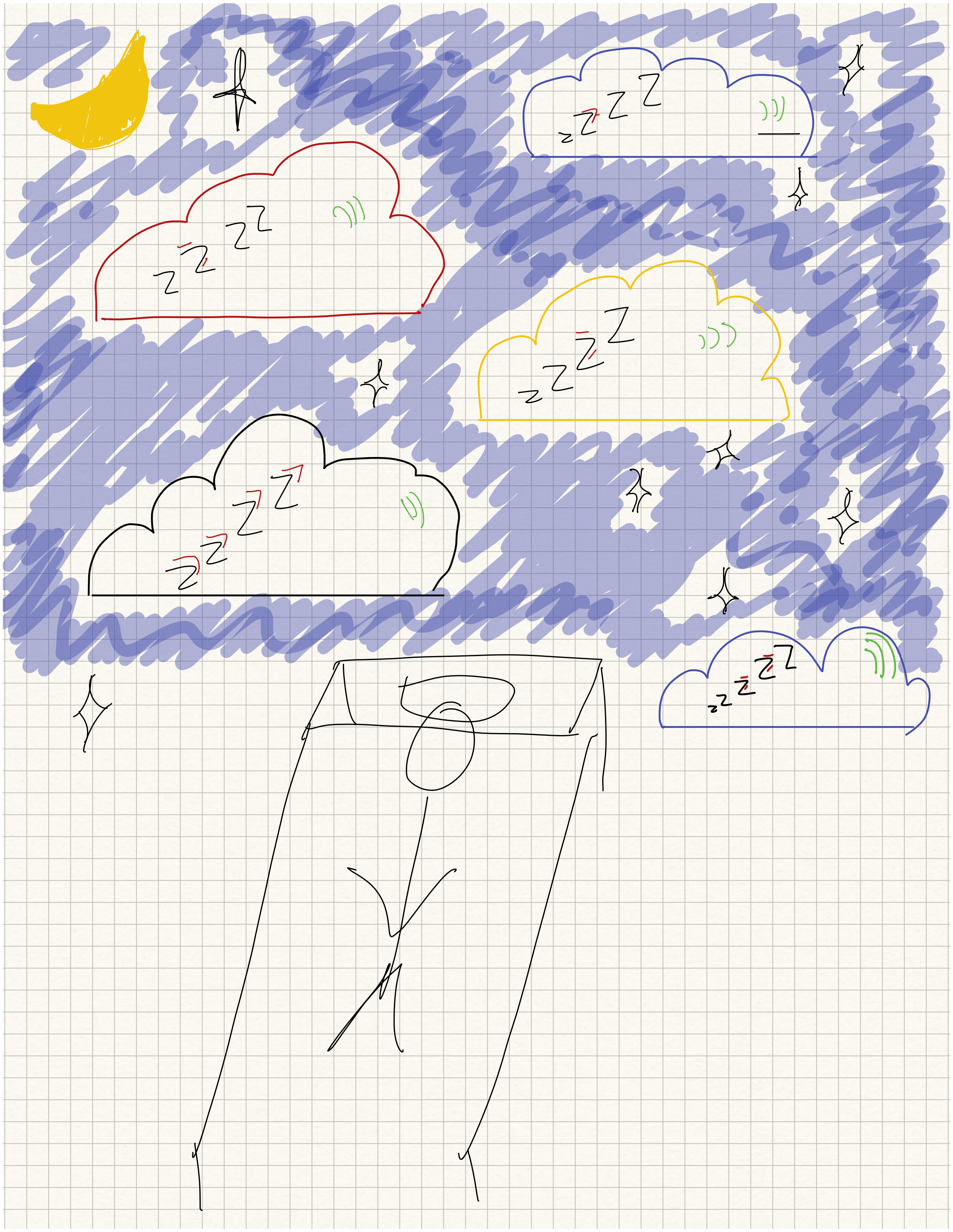
a. Describe in words what an affective visualization could look like for your personal data (3-5 sentences).

For my affective visualization, I am inspired by Stefanie Posavec and Giorgia Lupi's Dear Data project and make doodles surrounded by different aspects of sleep. I would love to draw a bed with someone sleeping at the bottom of the page and have little thought bubbles or clouds scattered throughout the paper to represent individual study sessions. Within these clouds, I want to include different doodles like Z's, lines, circles, and moons to represent different variables of these study sessions such as with friends or not, number of hours slept, caffeine or not, number of tasks completed, and houts spent on work.

b. Create a sketch (on paper) of your idea.



c. Make a draft of your visualization.



d. Write an artist statement

This drawing draws inspiration from Stefanie Posavec and Giorgia Lupi's Dear Data project as well as the drawings I see on Pinterest, and represents the productivity of work as a function of the amount of sleep I got the previous night. This piece is a watercolor painting of a night sky and person sleeping, with each cloud representing an individual study session. and doodles within each cloud representing different variables associated with each work session. The base layer will be water color and then overlaid with marker or color pencil to describe my work sessions.

e. Prep your materials to share in class

Slides: https://docs.google.com/presentati

on/d/1JiG_4HsGqKmk8n7m2uGnGQliuuB5ZrugnoLFuaByVXc/edit?usp=sharing

Problem 4. Statistical critique

a. Revisit and summarize

The authors performed numerous statistical tests to addresss the question, "Do humpback whale movements through Saldanha Bay follow the traditionally expected Southern Hemisphere migration patterns?". The test I focused on was a **Kruskal-Wallis** test, where the response variable is humpback whale group size and the predictor variable is seasons.

b. Visual clarity

The authors represented their statistics using a box plot-style figure to compare whale group size estimates across different seasons (Autumn to mid-winter - Mid- to late summer), showing which seasons were significantly different fand what seasons had higher recorded whale group size (mid-spring and early-summer). The x- and y- axes are in logical positions, with the season on the x-axis and best estimates for whale group size on the y-axis, the mean, SE, and sample size are well represented, as well as the comparisons from post-hoc tests. However, the y-axis title could be a little more clear including estimate instead of just "best group size" as it is hard to know without reading the figure caption. The underlying data is not represented in this figure.

c. Aesthetic clarity

I believe the authors handled “visual clutter” very well. The figure has minimal amount of grid lines, a white background, and no underlying data which in my opinion was a good decision as it would clutter the figure otherwise. The figure has a high data:ink ratio, the x- axis label is very well displayed, and the legend is clear and displayed well.

d. Recommendations

Regarding axis labels, I would suggest making the y- axis a little more clear and including “estimate” and “Humpback” so it is very clear to the reader on what they are looking at without having to read part of the paper or caption. Additionally, adding one more component to the legend at the top about the significant difference between seasons to once again make it more clear to the reader about what they are looking at. The sample sizes for each season were included in parentheses however it is not clear that they are sample sizes without looking at the figure caption, so either including n or adding () in the legend could make it more clear to the reader.